

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50

Annexure A

Code of Conduct:

I **P.Paneendra(192321072)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Paneendra

Annexure A

Scenario-based

[1] Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

[A] Explain why this issue occurs.

[B] Suggest a more secure block cipher mode and justify your choice based on the scenario.

ANS: Secure Data Transmission using Block Cipher

A. Why does ECB mode show patterns?

- In **Electronic Codebook (ECB)** mode, identical plaintext blocks always generate the same ciphertext blocks.
- When repeated financial information is encrypted, the resulting ciphertext also contains repeated patterns.
- These recurring patterns may provide clues to attackers, enabling them to analyze and estimate the structure of the original data even without access to the encryption key.

B. More secure block cipher mode

Recommendation: CBC (Cipher Block Chaining) mode (or GCM if authentication is also required).

Justification:

- CBC uses an Initialization Vector (IV) and combines each plaintext block with the previous ciphertext block before encryption.
- Identical plaintext blocks generate different ciphertext blocks.
- This hides data patterns and provides much better confidentiality than ECB.

[2] Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

[A] Identify the most suitable algorithm for this scenario.

[B] Justify your choice based on security strength and efficiency.

ANS: Choosing an Encryption Algorithm for Banking System.

A. Most suitable algorithm - Blowfish

B. Justification -

- DES: Not secure because it uses a 56-bit key, which can be broken by brute-force attacks.
- 3-DES: More secure than DES but is much slower due to triple encryption.
- Blowfish: Uses a 32–448-bit key, offers strong security, and performs faster than 3-DES.

Conclusion: Blowfish provides a good balance between high security and moderate performance for banking transactions.

[3] Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

[A] Explain how the key exchange works in this scenario.

[B] Identify a possible security threat and suggest a solution to mitigate it.

ANS: Key Exchange in an Unsecured Network

A. How Diffie–Hellman works

1. Both users publicly agree on a large prime number (p) and a generator (g).

2. User A chooses a private key and computes a public value.
3. User B chooses a private key and computes another public value.
4. They exchange their public values over the network.
5. Each user calculates the same shared secret using their own private key and the received public value.
6. The shared secret becomes the encryption key for secure communication.

B. Security threat and solution

Threat: Man-in-the-Middle (MITM) attack

- An attacker intercepts the exchanged public keys and establishes separate keys with each user.

Solution:

- Use digital certificates and digital signatures to authenticate users.
- Implement authenticated Diffie–Hellman (e.g., within TLS).

[4] Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

[A] Describe how RSA can be used for encryption and decryption in this scenario.

[B] Explain how key distribution and management play a role in maintaining security.

ANS: Public Key Cryptography for Email Security

A. RSA encryption and decryption

- The sender encrypts the email using the receiver's public key.
- Only the receiver can decrypt it using their private key.
- For authentication, the sender signs the message with their private key.
- The receiver verifies the signature using the sender's public key.

B. Key distribution and management

- Public keys must be distributed securely using Certificate Authorities (CAs).
- Private keys must remain secret and securely stored.
- Regular key updates, revocation, and certificate management help maintain security.

[5] Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication.

The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

[A] Analyze the constraints of IoT devices.

[B] Recommend a suitable cryptographic approach and justify your answer.

ANS: Cryptography for IoT Devices

A. Constraints of IoT devices

- Limited processing power
- Low memory capacity
- Limited battery life
- Low storage space
- Low communication bandwidth

B. Recommended cryptographic approach

Recommendation: Elliptic Curve Cryptography (ECC)

Justification:

- ECC provides security equivalent to RSA with much smaller key sizes.
- Requires less computation, memory, and power.
- Faster for low-power devices.
- Ideal for smart home and other IoT applications where resources are limited.

Example: A 256-bit ECC key provides security comparable to a 3072-bit RSA key while consuming significantly fewer resources.